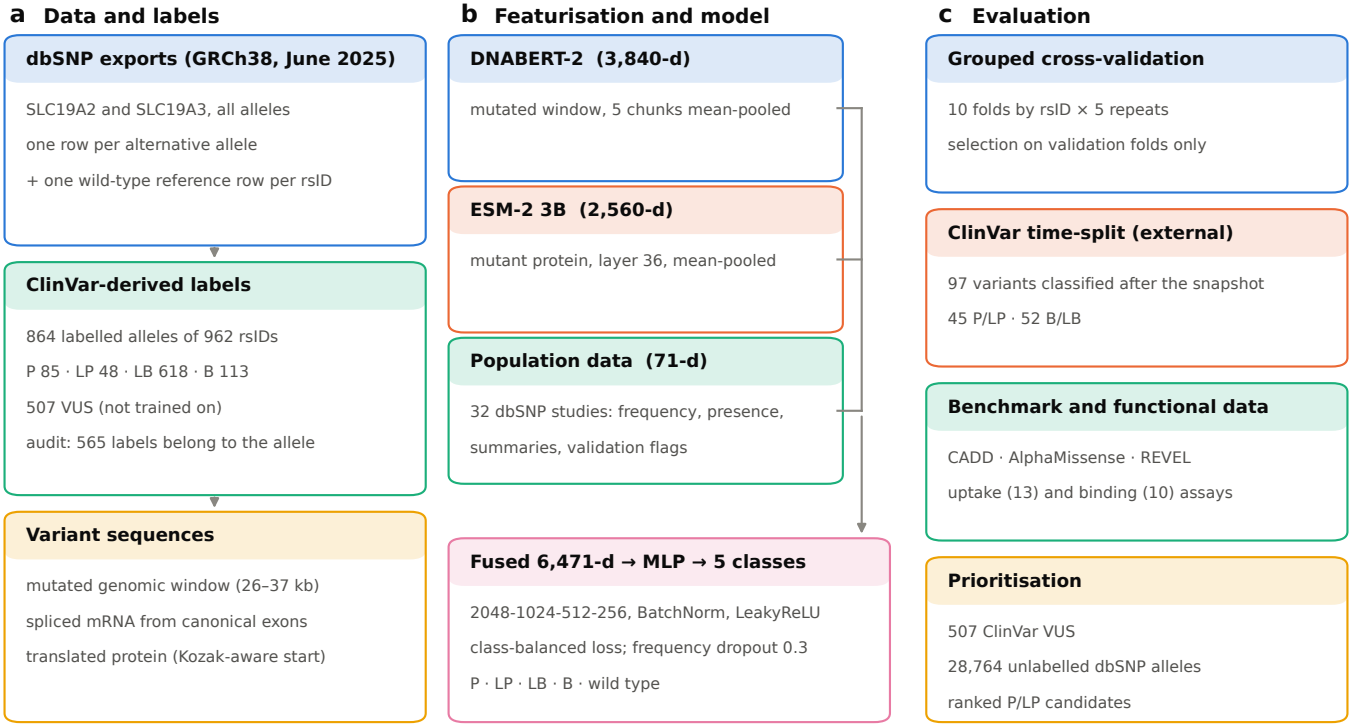
a Data and labels
dbSNP exports (GRCh38, June 2025)
SLC19A2 and SLC19A3, all alleles
one row per alternative allele
+ one wild-type reference row per rsID
ClinVar-derived labels
864 labelled alleles of 962 rsIDs
P 85 · LP 48 · LB 618 · B 113
507 VUS (not trained on)
audit: 565 labels belong to the allele
Variant sequences
mutated genomic window (26–37 kb)
spliced mRNA from canonical exons
translated protein (Kozak-aware start)
b Featurisation and model
DNABERT-2 (3,840-d)
mutated window, 5 chunks mean-pooled
ESM-2 3B (2,560-d)
mutant protein, layer 36, mean-pooled
Population data (71-d)
32 dbSNP studies: frequency, presence,
summaries, validation flags
Fused 6,471-d → MLP → 5 classes
2048-1024-512-256, BatchNorm, LeakyReLU
class-balanced loss; frequency dropout 0.3
P · LP · LB · B · wild type
c Evaluation
Grouped cross-validation
10 folds by rsID × 5 repeats
selection on validation folds only
ClinVar time-split (external)
97 variants classified after the snapshot
45 P/LP · 52 B/LB
Benchmark and functional data
CADD · AlphaMissense · REVEL
uptake (13) and binding (10) assays
Prioritisation
507 ClinVar VUS
28,764 unlabelled dbSNP alleles
ranked P/LP candidates